

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

The logs show that: There were many SYN, SYN/ACK packets sent to the server in a short period of time.

This event could be: SYN Flood

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. SYN (The client sends a SYN packet to the server to request a connection and synchronize sequence numbers)
2. SYN ACK (The server responds with a SYN-ACK packet, acknowledging the client's request and sending its own synchronization request)
3. ACK (The client sends an ACK packet back to the server to confirm the connection. After this step, the TCP connection is established and data transfer can begin.)

Explain what happens when a malicious actor sends a large number of SYN packets all at once: He makes a SYN Flood attack, which results in DoS

Explain what the logs indicate and how that affects the server: The server received an unusually large number of TCP SYN packets. This overloads server resources and prevents legitimate connections, which can cause service disruption or server failure